Group I, Row V, part 2b: UV term (dim. reg., $\overline{\text{MS}}$) + finite term, with $d_\perp$ kept general

This note does the following:

- It writes your part 2b as a UV-divergent term plus a finite term.
- The divergence at $x \to z$ is regulated in $d = 2 - \epsilon$ transverse dimensions in $\overline{\text{MS}}$, as in Group I Row II and Row IV.
- The polarization trace $\delta^{ii} = d_\perp$ of your first bracket is kept general. Setting $d_\perp = 2 - \epsilon$ reproduces the result in `Part2b_MSbar.pdf` and `Group_I_corrected.pdf`.
- Nothing is expanded in Λ/k^+.

1 Result

$$\boxed{\frac{d^3N_{2b}}{d^3k} = \frac{d^3N_{2b}}{d^3k}\bigg|_{\text{UV}} + \frac{d^3N_{2b}}{d^3k}\bigg|_{\text{finite}} \qquad (+\text{ c.c.})} \tag{1.1}$$

First term: UV divergent.

$$\begin{aligned}
\frac{d^3N_{2b}}{d^3k}\bigg|_{\text{UV}} = {}& \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2} \\
& \times \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[U^{af}(y) - U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y) \\
& \times \Bigg\{\Big[2\log\frac{\vee}{\Lambda} - 2\log\frac{\vee}{k^+} + 2\log\Big(1-\frac{\Lambda}{k^+}\Big) - 2 + \frac{4\Lambda}{k^+}\Big]\Big[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-z)^2\mu_{\overline{MS}}^2}{4}\Big] \\
& \quad + \frac{d_\perp}{2}\Big[\frac{1}{6} - \frac{\Lambda^2}{k^{+2}} + \frac{2\Lambda^3}{3k^{+3}}\Big]\Big[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-z)^2\mu_{\overline{MS}}^2}{4} + 1\Big]\Bigg\}.
\end{aligned} \tag{1.2}$$

- The pole is UV: $\epsilon > 0$.
- The "+1" in the second line comes from $\frac{d_\perp}{d} = \frac{d_\perp}{2}\big(1 + \frac{\epsilon}{2}\big)$ multiplying $\frac{2}{\epsilon}$. Here $d = 2 - \epsilon$ is the dimension of the x integral.
- The N_c of your colour factor $[N_cU(y) - N_cU(x)]$ is in the prefactor.

With $d_\perp = 2 - \epsilon$ (conventional dimensional regularization, as in Row II and Row IV), $\frac{d_\perp}{2}\big[\frac{2}{\epsilon} + \cdots + 1\big] = \frac{2}{\epsilon} + \ldots$, and $-2 + \frac{1}{6} = -\frac{11}{6}$:

$$\begin{aligned}
\frac{d^3N_{2b}}{d^3k}\bigg|_{\text{UV}} = {}& \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\Big[2\log\frac{\vee}{\Lambda} - 2\log\frac{\vee}{k^+} + 2\log\Big(1-\frac{\Lambda}{k^+}\Big) - \frac{11}{6} + \frac{4\Lambda}{k^+} - \frac{\Lambda^2}{k^{+2}} + \frac{2\Lambda^3}{3k^{+3}}\Big] \\
& \times \int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\Big[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-z)^2\mu_{\overline{MS}}^2}{4}\Big]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2} \\
& \times \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[U^{af}(y) - U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y)
\end{aligned} \tag{1.3}$$

(`sympy`: difference from Eq. (1.2) at $d_\perp = 2 - \epsilon = 0$).

$\overline{\text{MS}}$. The factor $\big(\frac{\mu^2 e^{\gamma_E}}{4\pi}\big)^{\epsilon/2}$ of the x integral is already in $\mu_{\overline{MS}}$, so $\overline{\text{MS}}$ removes only $\frac{2}{\epsilon}$:

$$\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-z)^2\mu_{\overline{MS}}^2}{4} \;\longrightarrow\; 2\gamma_E + \log\frac{(y-z)^2\mu_{\overline{MS}}^2}{4} = \log\frac{(y-z)^2\mu_{\overline{MS}}^2 e^{2\gamma_E}}{4}. \tag{1.4}$$

Second term: finite ($d = 2$; $d_\perp$ as in your first bracket):

$$\left.\frac{d^3N_{2b}}{d^3k}\right|_{\text{finite}} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\Bigg\{$$

$$e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$

$$\times\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+(k^+-p^+)}{k^{+2}} - \frac{p^+}{k^+-p^+}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2} - \frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]$$

$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]$$

$$-\frac{k^+-p^+}{p^+}\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]$$

$$+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]$$

$$-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\Bigg]$$

$$\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)$$

$$-\frac{1}{k^+}\left[\frac{d_\perp}{2}\frac{p^+(k^+-p^+)}{k^{+2}}+\frac{p^+}{k^+-p^+}+\frac{k^+-p^+}{p^+}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}$$

$$\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y)\Bigg\}. \tag{1.5}$$

The first six lines inside the braces are your 2b, unchanged. The last two lines subtract the counterterm: no phase $e^{-ik\cdot(z-x)p^+/k^+}$, colour $[N_cU(y)-N_cU(z)]$, kernel $\frac{(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2}$. The finite term is UV finite and is evaluated in two dimensions. Its p^+ integral is over $\Lambda < p^+ < k^+-\Lambda$ with no plus prescription. For $d_\perp = 2$ it is Eq. (1.5) of `Part2b_MSbar.pdf`.

2 Derivation

Notation: $\xi \equiv \frac{p^+}{k^+}$, $\mathbf{r} \equiv x - z$, $Y \equiv y - z$, and $K(a,b,c) \equiv \frac{(a-c)\cdot(b-c)}{(a-c)^2(b-c)^2}$ (the third argument is integrated over).

Step 1: where it diverges

At $x \to z$ the integrand behaves like $\frac{1}{(x-z)^2}$. The colour factor $N_c[U(y)-U(x)] \to N_c[U(y)-U(z)]$ does not vanish there, so $\int d^2x$ diverges logarithmically.

The other regions are finite:

- $y \to x$ and $y \to z$ are at most $\frac{1}{|y-x|}$ or $\frac{1}{|y-z|}$;
- at x, y, z all together the integrand scales as ρ^{-3}, against $d^2x\,d^2y \sim \rho^3 d\rho$;
- at large $|x-z|$ the x integral converges through the phase $e^{-ik\cdot(z-x)p^+/k^+}$.

Step 2: the short-distance limit (with $d_\perp \neq d$)

Set $z = 0$ and $x = \mathbf{r} \to 0$. Then:

- the phase $\to 1$;
- $\frac{1}{p^+(y-x)^2+(k^+-p^+)(y-z)^2} \to \frac{1}{k^+(y-z)^2}$;
- $\frac{(z-x)^m}{(z-x)^2} = -\frac{r^m}{r^2}$.

Average over the directions of $\mathbf{r}$ in d dimensions, using $\langle r^i r^j\rangle = \frac{r^2}{d}\delta^{ij}$. Call your six brackets $B_1, \ldots, B_6$; they give:

term	p^+ factor	leading term of $-\frac{r^m}{r^2}B^{im}$	angular average
B_1	$\xi(1-\xi)$	$d_\perp \frac{r^i(Y\cdot r)}{r^4} - \frac{d_\perp}{2}\frac{r^i}{r^2}$	$\frac{d_\perp}{d}\frac{Y^i}{r^2}$
B_2	$-\frac{\xi}{1-\xi}$	$-\frac{Y^i}{r^2} + \frac{r^i}{r^2} + \frac{(Y\cdot r)Y^i}{2Y^2r^2}$	$-\frac{Y^i}{r^2}$
B_3	$1-\xi$	$-2\frac{r^i(Y\cdot r)}{r^4}$	$-\frac{2}{d}\frac{Y^i}{r^2}$
B_4	$-\frac{1-\xi}{\xi}$	$-\frac{Y^i}{r^2} - \frac{(Y\cdot r)Y^i}{2Y^2r^2}$	$-\frac{Y^i}{r^2}$
B_5	ξ	$-2\frac{r^i(Y\cdot r)}{r^4} + 2\frac{r^i}{r^2}$	$-\frac{2}{d}\frac{Y^i}{r^2}$
B_6	-1	$-2\frac{r^i(Y\cdot r)}{r^4} + \frac{r^i}{r^2}$	$-\frac{2}{d}\frac{Y^i}{r^2}$

With the p^+ factors:

$$\frac{d_\perp}{d}\,\xi(1-\xi) + \frac{\xi}{1-\xi} - \frac{2}{d}(1-\xi) + \frac{1-\xi}{\xi} - \frac{2}{d}\,\xi + \frac{2}{d} = \frac{d_\perp}{d}\,\xi(1-\xi) + \frac{\xi}{1-\xi} + \frac{1-\xi}{\xi}. \tag{2.1}$$

The $\frac{1}{d}$ terms of B_3, B_5, B_6 cancel. Only B_1 keeps $\frac{d_\perp}{d}$. Numerically, for $d = 2, 3$ and $d_\perp = 1, 2, 3, 5$: rel. 2×10^{-9}. So, at $x \to z$, your 2b is

$$\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\frac{dp^+}{2\pi}\,e^{-ik\cdot(w'-z)}\,\frac{1}{k^+}\Big[\frac{d_\perp}{d}\frac{p^+(k^+-p^+)}{k^{+2}} + \frac{p^+}{k^+-p^+} + \frac{k^+-p^+}{p^+}\Big]$$
$$\times\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\,\frac{1}{(x-z)^2}\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_cU^{af}(y) - N_cU^{af}(z)\Big]\rho^{b'}(x')\rho^f(y)$$

plus terms with zero angular average and $O(\frac{1}{|x-z|})$.

Step 3: the counterterm

Replace $\frac{1}{(x-z)^2}$ by

$$K(z,z,x) - K(y,z,x) = \frac{(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2}. \tag{2.2}$$

It has the following properties:

- the same $\frac{1}{(x-z)^2}$ at $x \to z$, up to a term with zero angular average;
- it falls like $|x-z|^{-3}$ at large $|x-z|$;
- in two dimensions its angular average is exactly $\frac{\theta(|y-z|-|x-z|)}{(x-z)^2}$ (checked: max. dev. 9×10^{-16}).

The counterterm CT is Step 2 with this kernel, integrated over $\Lambda < p^+ < k^+ - \Lambda$. Then

- $\frac{d^3N_{2b}}{d^3k} = (2b - \mathrm{CT}) + \mathrm{CT}$;
- CT, done in d dimensions, is the UV term;
- $2b - \mathrm{CT}$ at $d = 2$ is the finite term. In it the $\frac{d_\perp}{d}$ becomes $\frac{d_\perp}{2}$, Eq. (1.5).

Step 4: the x integral in $d = 2 - \epsilon$, $\overline{\mathrm{MS}}$

In d dimensions each kernel $\frac{r^m}{r^2}$ becomes $c_d\frac{r^m}{(r^2)^{d/2}}$, with $c_d = \Gamma(\frac{d}{2})\pi^{1-\frac{d}{2}}$, the Fourier transform of $-2\pi i\frac{q^m}{q^2}$.

- $\int d^dx\, K_d(z,z,x) = 0$ (scaleless).
- $\int d^dx\, K_d(y,z,x) = 4\pi^2\int\frac{d^dq}{(2\pi)^d}\frac{e^{iq\cdot(y-z)}}{q^2} = \pi^{2-\frac{d}{2}}\Gamma(\frac{d}{2}-1)[(y-z)^2]^{1-\frac{d}{2}}$.

With $\big(\frac{\mu^2e^{\gamma_E}}{4\pi}\big)^{\epsilon/2}$,

$$\int d^dx\,\big[K_d(z,z,x) - K_d(y,z,x)\big] = -\pi\,\Gamma\Big(-\frac{\epsilon}{2}\Big)\Big(\frac{(y-z)^2\mu^2e^{\gamma_E}}{4}\Big)^{\frac{\epsilon}{2}} = \pi\Big[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\Big] + O(\epsilon). \tag{2.3}$$

The same result follows from the sharp cut $\frac{\theta(|y-z|-|x-z|)}{(x-z)^2}$ and from the smooth form $\frac{1}{(x-z)^2} - \frac{1}{(x-z)^2+(y-z)^2}$ (`sympy`: all 0).

Step 5: the p^+ integral (exact)

With $\lambda = \frac{\Lambda}{k^+}$ and $\int_\Lambda^{k^+-\Lambda} \frac{dp^+}{k^+} = \int_\lambda^{1-\lambda} d\xi$:

$$\int_\lambda^{1-\lambda} d\xi \Big[\frac{d_\perp}{d}\xi(1-\xi) + \frac{\xi}{1-\xi} + \frac{1-\xi}{\xi}\Big] = \frac{d_\perp}{d}\Big[\frac{1}{6} - \lambda^2 + \frac{2}{3}\lambda^3\Big] + 2\log\frac{1-\lambda}{\lambda} - 2 + 4\lambda \tag{2.4}$$

(numerically, rel. 2×10^{-15}), and $2\log\frac{1-\lambda}{\lambda} = 2\log\frac{\vee}{\Lambda} - 2\log\frac{\vee}{k^+} + 2\log(1-\frac{\Lambda}{k^+})$.

The pole multiplies $\frac{d_\perp}{d}$. With $d = 2-\epsilon$,

$$\frac{d_\perp}{d}\Big[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-z)^2\mu^2}{4}\Big] = \frac{d_\perp}{2}\Big[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-z)^2\mu^2}{4} + 1\Big] + O(\epsilon)$$

(`sympy`: 0). This is the second line of Eq. (1.2).

Step 6: prefactor

$\frac{g^4}{8\pi^4}\cdot\frac{1}{k^+}\cdot\frac{1}{2\pi}\cdot\pi\cdot N_c = \frac{g^4 N_c}{16\pi^4}\frac{1}{k^+}$ (`sympy`: 0). Here:

- $\frac{1}{2\pi}$ is from $\frac{dp^+}{2\pi}$;
- one $\frac{1}{k^+}$ of Step 2 turns dp^+ into $d\xi$;
- π is from Eq. (2.3);
- N_c is from the colour factor.

Relative to LO, $\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}$, the factor is $\frac{\alpha_s N_c}{2\pi}$. The integral is the LO one with the charge at y and the gluon at z.

Step 7: the finite term is finite

At $x\to z$ the $\frac{1}{(x-z)^2}$ terms cancel between 2b and CT for every p^+. Numerically, with a smooth test colour factor and the phase kept, $r^2\langle 2b - \mathrm{CT}\rangle$ at $r = 10^{-2} \to 10^{-3}$ falls from 1.5×10^{-4} to 1.5×10^{-6} ($d_\perp = 2, 3$; $\xi = 0.02, 0.37, 0.95$). Over the same range $r^2\langle 2b\rangle$ stays at 0.09–1.75. At large $|x-z|$ the counterterm falls like $|x-z|^{-3}$, so the finite term behaves there like 2b.

3 Checks

Scripts `part2b_dperp_checks.py` (new) and `part2b_checks.py` (`Part2b_MSbar.pdf`).

check	result
short-distance coefficient (2.1), $d = 2, 3$, $d_\perp = 1, 2, 3, 5$	rel. 2×10^{-9}
angular average of the counterterm kernel $= \frac{\theta(\lvert y-z\rvert-\lvert x-z\rvert)}{(x-z)^2}$	max. dev. 9×10^{-16}
x integral in $d = 2-\epsilon$: K form, sharp cut, smooth form (`sympy`)	all 0
p^+ integral with $\frac{d_\perp}{d}$	rel. 2×10^{-15}
$\frac{d_\perp}{d}$ expansion (`sympy`)	0
$d_\perp = 2-\epsilon$ gives Eq. (1.3) (`sympy`)	0
prefactor (`sympy`)	0
finite term: $r^2\langle 2b - \mathrm{CT}\rangle \to 0$	$1.5\times10^{-4} \to 1.5\times10^{-6}$